

leetcode.com/problems/maximum-subarray/

Continue where you left off: Chrome can restore your tabs every time you restart. To turn this on, go to [Settings](#).

Problem List

53. Maximum Subarray

Solved

Description

Editorial

Submissions

Solutions

Medium

Topics

Companies

Given an integer array `nums`, find the **subarray** with the largest sum, and return *its sum*.

Example 1:

Input: `nums = [-2,1,-3,4,-1,2,1,-5,4]`

Output: 6

Explanation: The subarray `[4,-1,2,1]` has the largest sum 6.

Example 2:

Input: `nums = [1]`

Output: 1

Explanation: The subarray `[1]` has the largest sum 1.

Example 3:

Input: `nums = [5,4,-1,7,8]`

Output: 23

Explanation: The subarray `[5,4,-1,7,8]` has the largest sum 23.

Constraints:

1

$\leq$

nums.length

$\leq$

10⁵

-10⁴

$\leq$

nums[i]

$\leq$

10⁴

Follow up:

If you have figured out the `O(n)` solution, try coding another solution using the **divide and conquer** approach, which is more subtle.

38.4K

592

328 Online

Code

Java

Auto

```
1 class Solution {
2     public int maxSubArray(int[] nums) {
3
4         int currentSum = nums[0];
5         int maxSum = nums[0];
6
7         for (int i = 1; i < nums.length; i++) {
8
9             currentSum = Math.max(nums[i], currentSum + nums[i]);
10
11             maxSum = Math.max(maxSum, currentSum);
12         }
13
14         return maxSum;
15     }
16 }
```

Saved

Ln 16, Col 2

Testcase

Test Result

Accepted

Runtime: 0 ms

Case 1

Case 2

Case 3

Input

nums =

`[-2,1,-3,4,-1,2,1,-5,4]`

Output

6

Expected

6

Contribute a testcase